Underdetermined (Deterministic) Problems

Consider $y = Ax$ where

$y \in \mathbb{R}^{\cancel{p}\,m}$, $x \in \mathbb{R}^n$, $n > \cancel{p}\,m$. The

equations are consistent if $\exists\ \forall y \in \mathbb{R}^m$,

$\exists\ x_0 \in \mathbb{R}^n$ s.t. $y = Ax_0$,

We define the set of solutions

$$V = \{x \in \mathbb{R}^n \mid y = Ax\} \neq \emptyset$$

if consistent. If we define

$\mathcal{N}(A) = \{x \in \mathbb{R}^n \mid Ax = 0\}$ = null space of A, then clearly, if x_0 is one solution of the equations, then $\forall \bar{x} \in \mathcal{N}(A)$, $x_0 + \bar{x}$ is also a solution, that is

$$y = A(x_0 + \bar{x})$$

<u>Exercise</u> (Easy) If $y = Ax_0$, then

$$V = \{x \in \mathbb{R}^n \mid y = Ax\} = x_0 + \mathcal{N}(A)$$

$\square$

How to choose among the set of solutions?

Common choice: take the one of minimum norm

$$\hat{x} = \arg\min_{x \in V} \|x\|$$

Here is the general version of the problem

Let $(\mathcal{X}, \mathcal{F}, \langle \cdot, \cdot \rangle)$ be an inner product space, let $\{y_1, \ldots, y_m\}$ be a linearly independent set, and let $c_1, \ldots, c_m \in \mathcal{F}$ be a set of scalars. Define

$$V = \{x \in \mathcal{X} \mid \langle x, y_i \rangle = c_i,\ i = 1, \ldots, m\}$$

Exercise: $\exists !\ x_0 \in \operatorname{span}\{y_1, \ldots, y_m\}$ such that $\langle x_0, y_i \rangle = c_i$, $i = 1, \ldots, m$

Remark: Equivalent: $\underbrace{}_{}$ $\exists !\, x_0 \in \mathcal{X}$ s.t. $V \cap \operatorname{span}\{y_1, \ldots, y_m\} = \{x_0\}$

$\square$

Solution: Write $x_0 = \beta_1 y_1 + \cdots \beta_m y_m$ obtain the normal equations, and deduce that the Gram matrix is nonsingular. □

Notation: $[y_1, \ldots, y_m] := \text{span}\{y_1, \ldots, y_m\}$.

Proposition $V = x_0 + [y_1, \ldots, y_m]^\perp$

Proof. ~~To show:~~

To show: $x \in V \iff x - x_0 \perp [y_1, \ldots, y_m]$

$$
\begin{aligned}
x \in V &\iff \langle x, y_i \rangle = c_i \quad i = 1, \ldots, m \\
&\iff \langle x - x_0, y_i \rangle = \langle x, y_i \rangle - \langle x_0, y_i \rangle \\
&\qquad\qquad = c_i - c_i = 0 \quad i = 1, \ldots, m \\
&\iff (x - x_0) \perp [y_1, \ldots, y_p]
\end{aligned}
$$
□

<u>Thm</u> Let $(\mathcal{X}, \mathcal{F}, \langle\cdot,\cdot\rangle)$ be a finite dimensional inner product space and $\{y_1, \dots, y_m\}$ a linearly indep. set, $c_1, \dots, c_m$ a set of scalars. Then $\exists!\, v^* \in V = \{x \in \mathcal{X} \mid \langle x, y_i\rangle = c_i,\ i = 1, \dots, m\}$ having minimum norm, and v^* is characterized by $v^* \perp [y_1, \dots, y_p]^\perp$ (not $v^* \perp V$).

<u>Proof:</u> Let $M = [y_1, \dots, y_p]^\perp$ and write

$$V = x_0 + M.$$

The problem is to determine

$$\inf_{v \in V} \|v\| = \inf_{m \in M} \|x_0 + m\| = \inf_{m \in M} \|x_0 - m\|$$

because M is a subspace. By the Projection Thm, $\exists!\, x^* \in M$ such that

$$\|x_0 - x^*\| = \inf_{m \in M} \|x_0 - m\|, \text{ and}$$

x^* is characterized by $x_0 - x^* \perp M$.

We set $v^* = x_0 - x^*$, and then

$$\|v^*\| = \inf_{v \in V} \|v\| \quad \text{and}$$

$v^* \perp M$.

$\square$

<u>Remark</u> $v^* \perp [y_1, \ldots, y_p]^\perp \iff$

$v^* \in \left([y_1, \ldots, y_p]^\perp \right)^\perp \iff v^* \in [y_1, \ldots, y_p]$

<u>Because</u> $\mathcal{X} = [y_1, \ldots, y_p] \oplus [y_1, \ldots, y_p]^\perp$

$\triangle$

<u>Thm</u> Let $(\mathcal{X}, \mathcal{F}, \langle \cdot, \cdot \rangle)$ be an inner product space, $\{y_1, \ldots, y_m\}$ a linearly independent set, and $c_1, \ldots, c_m$ constants. Let

$$V = \{ x \in X \mid \langle x, y_i \rangle = c_i, \ i = 1, \ldots, m \}$$

Then $\exists !\ v^* \in V$ s.t. $\|v^*\| = \inf_{v \in V} \|v\|$.

Moreover, $v^* = \sum_{i=1}^{m} \beta_i y_i$

Where the β_i's satisfy the normal equations:

$$\begin{bmatrix} \langle y_1, y_1\rangle & \langle y_2, y_1\rangle & \cdots & \langle y_p, y_p\rangle \\ \langle y_1, y_m\rangle & \langle y_2, y_m\rangle & \cdots & \langle y_p, y_m\rangle \end{bmatrix} \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_m \end{bmatrix} = \begin{bmatrix} c_1 \\ \vdots \\ c_m \end{bmatrix}$$

Proof. From our remark, $v^* \in [y_1, \ldots, y_m]$. We write

$v^* = \beta_1 y_1 + \cdots + \beta_p y_m$ and then impose $v^* \in V$. This gives the normal equations.

$\square$

Return to $y = Ax$

$y \in \mathbb{R}^m$, $x \in \mathbb{R}^n$, $n > m$, $y = Ax$.

We define an inner product on $\mathbb{R}^n$ by $\langle x, \tilde{x} \rangle = x^T Q \tilde{x}$, $Q > 0$.

So that $\|x\| = \left(x^T Q x \right)^{1/2}$.

<u>Thm</u> If the rows of A are linearly independent, then

$$\hat{x} = \underset{Ax=b}{\arg\min} \|x\|$$

is given by

$$\boxed{\hat{x} = Q^{-1} A^T \left(A Q^{-1} A^T \right)^{-1} b}$$

Proof

We write $A = \begin{bmatrix} a_1 \\ \vdots \\ a_m \end{bmatrix}$ and $\tilde{A} = AQ^{-1}$

define $v_i \in \mathbb{R}^n$ by

$$v_i = (a_i Q^{-1})^T = Q^{-1} a_i^T$$

because Q is symmetric. It follows that

$$Ax = b \Leftrightarrow \langle v_i , x \rangle = b_i \quad i = 1, .., m$$

Hence,

$$\hat{x} = x^* = \sum_{i=1}^{m} \beta_i v_i = [v_1 | \cdots | v_m] \beta$$

and

$$G\beta = b$$

and G is the Gram matrix.

Exercise

$$G = AQ^{-1}A^T \quad \text{and} \quad G = G^T$$

$$[v_1 | \cdots | v_m] = Q^{-1}A^T$$

$\square$

Hence, $\hat{x} = Q^{-1}A^T (AQ^{-1}A^T)^{-1} b$

$\square$

Sol. to Exercise

$$v_i = Q^{-1} a_i^T$$

$$[v_1 | \cdots | v_m] = [Q^{-1}a_1^T | \cdots | Q^{-1}a_m^T]$$

$$= Q^{-1}[a_1^T | \cdots | a_m^T]$$

$$= Q^{-1}A^T$$

$$
\begin{aligned}
G_{ij} &= \langle v_i , v_j \rangle \\
&= v_i^T Q v_j \\
&= (a_i Q^{-1}) Q Q^{-1} a_j^T \\
&= a_i Q^{-1} a_j^T
\end{aligned}
$$

$$
\begin{aligned}
(A Q^{-1} A^T)_{ij} &= a_i \left(Q^{-1} A^T\right)_{j\text{-column}} \\
&= a_i Q^{-1} a_j^T
\end{aligned}
$$

$\square$